FlowRecon-Bench: A benchmark for sparse-sensor flow-field reconstruction across flow regimes, with calibrated uncertainty

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Reconstructing a full flow field from sparse point sensors is a core task of experimental and operational fluid mechanics, and a rapidly growing machine-learning literature now addresses it with deterministic operator learners and conditional generative models. Yet the field has no shared benchmark: published methods are evaluated on incompatible, often temporally leaky splits, ensembles are drawn but never scored with proper probabilistic metrics, and classical baselines are rarely reported. We present FlowRecon-Bench, a benchmark spanning three flow regimes of increasing dimensionality and dynamical complexity — laminar cylinder vortex shedding (2D, multi-Reynolds), two-dimensional Kolmogorov turbulence, and three-dimensional isotropic turbulence (JHTDB) — evaluated with leakage-audited splits, frozen sensor manifests, per-channel accuracy, proper scoring rules (CRPS, coverage, spread–error), physics metrics (windowed spectra, structure functions), and matched cost accounting (training and inference runtime and peak memory) for every method. Across a fidelity-audited fleet of twelve-plus methods — interpolation and gappy-POD anchors, deterministic operator learners, latent and ambient generative models including DMF-Gen — we find that method ranking is regime-dependent, and no method wins twice: gappy POD with 20 modes wins the laminar cylinder outright (0.056 relative L_2, including the never-observed pressure at 0.051), a deterministic operator learner leads two-dimensional turbulence (Geo-FNO 0.417) while being beaten on CRPS by three of the four generative models, and latent flow matching, SiT and interpolation split the channels of 3D turbulence. No method recovers unobserved channels of 3D turbulence, though gappy POD recovers them on the cylinder; nearly every scored ensemble is under-dispersed, and post-hoc recalibration partially repairs it. Benchmark scores are systematically worse than literature-reported numbers on the same data, and we show by controlled ablation that the gap is evaluation protocol — sensor layout, split design, and reporting convention — rather than model capability; the same checkpoint improves by 25% when its scattered sensors are rearranged onto a uniform lattice, while split leakage proves severe in the small-data 3D regime and negligible at 2D data scales. We release the datasets, splits, sensor manifests, configurations, and evaluation driver as a versioned artifact. [TODO: finalize abstract numbers and artifact DOI]

I. INTRODUCTION

Flow fields govern systems that experiments can only observe pointwise: particle-image velocimetry resolves planes of a turbulent flow while the surrounding volume goes unmeasured, flight-test campaigns record surface pressure at taps while the off-body flow that determines performance is inaccessible, and environmental monitoring networks sample the atmosphere at stations kilometers apart. Recovering the full field from such sparse, indirect data is a long-standing inverse problem,[1–3] and at realistic sensor densities it is severely ill-posed: many fields fit the data, and the missing information sits disproportionately at the small scales. A machine-learning literature now more than a decade deep addresses it with deterministic reconstruction networks,[4–7] and, more recently, with conditional generative models that sample the distribution of fields consistent with the observations rather than regressing to its mean.[8–13]

What that literature does not have is a shared evaluation. Community benchmarks for scientific machine learning — PDEBench, The Well, CFDBench, AirfRANS, and their successors — pose forward-surrogate or uniform-downsampling super-resolution tasks, not reconstruction from scattered point sensors; the Johns Hopkins Turbulence Database is a query service with no canonical protocol, and published sparse-reconstruction papers each cut their own incompatible extraction from it. Three consequences follow, and this paper documents all three. First, *splits leak*: turbulence snapshots are strongly autocorrelated in time, and the standard random-in-time split makes every test frame a near-duplicate of a training frame. On our 3D turbulence data, frame-to-frame correlation is 1.00 at unit separation; under a shuffled split the strongest latent generative baseline scores 0.14 relative L_2, and under an honest spatially disjoint split the same model scores 0.56 — a factor of four, larger than every method-to-method gap we measure (Sec. V F). Second, *ensembles are drawn but not scored*: generative reconstruction papers routinely show mean-and-standard-deviation maps, several claim calibrated uncertainty on that evidence alone, and, to our knowledge, none reports a proper scoring rule or a calibration diagnostic for 3D single-snapshot reconstruction from scattered sensors;[14–18] every scored precedent in adjacent settings is under-dispersed. Third, *classical baselines are absent*: training-free interpolation and gappy proper-orthogonal-decomposition floors are rarely reported, and we find one of them (inverse-distance weighting) beats every deep model's observed-channel error on 3D turbulence at 1% sensor density. A consequence worth stating plainly up front: every number and image in this paper will look worse than its counterparts in the literature, and Sec. V F shows by controlled ablation — same checkpoints, protocols swapped — that this gap is the protocol, not the models.

FlowRecon-Bench is our response: a benchmark for sparse-sensor flow reconstruction built to answer the question a practitioner actually has — *which method family should I use, in which flow regime, at what sensor budget, with what uncertainty and at what cost?* It spans three regimes forming a ladder in dimensionality and dynamical complexity (Sec. II B): laminar cylinder vortex shedding, where the dynamics are low-dimensional and classical linear methods should be near-optimal; two-dimensional Kolmogorov turbulence, the standard testbed of the 2D generative literature;[8,9] and three-dimensional isotropic turbulence under a spatially disjoint evaluation protocol, where sensors observe only two of four channels. Multiphysics and unstructured-geometry regimes are deferred to follow-up work; within this suite the cylinder's body-fitted mesh already carries a surface-confined sensing variant (Sec. IV B). Every method is evaluated under identical seeded sensor draws, identical augmentation, leakage-controlled splits with a strict tune/test separation, per-channel accuracy, fair CRPS, coverage and spread–error diagnostics, windowed spectral metrics, and — because model selection in practice is a cost–benefit decision — measured training and inference runtime and peak memory for every row.

Our contributions are the benchmark and the findings it forces. (i) A three-regime task suite with leakage-audited splits, frozen sensor manifests, and a fingerprinted evaluation driver, released as a versioned artifact together with per-method configurations and audit notes. (ii) A fidelity-audited fleet of twelve-plus methods spanning classical anchors, deterministic operator learners, latent generative and ambient generative routes, parameter-matched where the published configurations allow and budget-disclosed where they do not (Sec. III). (iii) Findings that generalize beyond any single table: method ranking is regime-dependent; no method recovers channels the sensors do not observe in 3D turbulence (a fleet-wide identifiability limit, not a per-method defect); accuracy and sample realism trade off through the sampler's operating point, so ranking generative models on mean-field error alone is misleading (Sec. V A); nearly every scored ensemble is under-dispersed, and a post-hoc recalibration fitted on a disjoint tuning split repairs stated coverage (Sec. V C); and the computational axis separates grid methods, whose memory walls we measure, from point methods, whose costs are flat in resolution (Sec. V E).

TABLE I. The three regimes of FlowRecon-Bench. Each step of the ladder adds one source of difficulty; the split column names the leakage-controlled protocol (Sec. II C).

Regime	Dataset	Points	Channels	Split	Axis isolated
2D laminar periodic	Cylinder wake (new)	2.4×10^4	u, v, p	cross-Re	low intrinsic dim.
2D chaotic	Kolmogorov 256[28]	6.6×10^4	ω	cross-traj.	2D multiscale
3D chaotic	JHTDB isotropic[19]	1.95×10^6	U_x, U_y, U_z, p	spatial	unobs. channels

II. BENCHMARK DESIGN

A. Task

Let $u : \Omega \to \mathbb{R}^C$ be a flow field drawn from a distribution over fields (snapshots of a turbulent flow, solutions over varying geometry). It is observed through a measurement operator returning readings $\mathcal{O} = \{(\boldsymbol{x}_j,\, c_j,\, y_j)\}_{j=1}^M$ with $y_j = u_{c_j}(\boldsymbol{x}_j) + \varepsilon_j$, where $\boldsymbol{x}_j$ is a sensor location, c_j indexes the observed quantity, and ε_j is noise. The task is single-snapshot reconstruction: estimate u — or, for probabilistic methods, sample an approximation to $p(u \mid \mathcal{O})$ — at all query locations, with no access to sensor time histories, initial conditions, or the governing solver. This covers sensors scattered arbitrarily or confined to a surface, observed quantities forming a subset of the target channels, and corrupted readings. Sequential assimilation and forward surrogacy are deliberately out of scope; they are different problems with different information budgets.

B. Datasets: three flow regimes

The three regimes form a ladder in both spatial dimension and dynamical complexity — 2D laminar and periodic, 2D chaotic, 3D chaotic — so that each step isolates one source of difficulty while holding the task fixed (Table I); preprocessing details and download/regeneration recipes ship with the artifact.

a. Cylinder wake (new). Two-dimensional incompressible flow past a circular cylinder, generated with OpenFOAM 9 (`pimpleFoam`, laminar) on a 12-block O-grid of 23,800 cells with boundary-layer grading (`checkMesh` clean; maximum non-orthogonality 43.9°). Nondimensionalized by the cylinder diameter and free-stream speed ($D = U_\infty = 1$; $\nu = 1/\mathrm{Re}$), at $\mathrm{Re} \in \{60, 80, 100, 150, 200, 250\}$. Each case runs a transient phase to $t = 120$ and a sampling phase to $t = 270$ written every 0.5 time units — 300 snapshots spanning roughly 25 shedding cycles at $\mathrm{Re} = 100$. Strouhal numbers of the generated cases are verified against established measurements[20,21] as the physics-level validation of the data: from the lift-coefficient spectrum over the sampling window, $\mathrm{St} = 0.140$, 0.160, 0.173, 0.187, 0.200, 0.207 at $\mathrm{Re} = 60$, 80, 100, 150, 200, 250, against the empirical $\mathrm{St}(60) \approx 0.135$, $\mathrm{St}(100) \approx 0.164$, $\mathrm{St}(150) \approx 0.183$, $\mathrm{St}(200) \approx 0.196$ — the 3–5% excess is the familiar two-dimensional overprediction of the shedding frequency, and the monotone trend and spacing are reproduced (`src/check_strouhal.py`). Mean drag lies between 1.40 and 1.46, consistent with the 6% blockage of the domain, and the lift-coefficient amplitude increases monotonically with Re (0.16 to 0.88). The dataset ships in two exports: the native cell-center point cloud (with the wall-adjacent cell ring indexed for a surface-tap sensing variant, the 2D analogue of surface-only sensing on a body) and a 400×200 uniform-grid interpolation over $x \in [-3, 17]$, $y \in [-5, 5]$ with a body mask, for grid-locked methods — the need for the second export is itself a capability observation. Cross-Re protocol: train on $\mathrm{Re} \in \{60, 100, 150, 200\}$, evaluate on the held-out $\mathrm{Re} \in \{80, 250\}$ (one interpolative, one mildly extrapolative), with temporally blocked splits inside the training cases. An observe-u-only variant probes cross-channel inference (v, p unobserved) in a regime where, unlike 3D turbulence, the phase of the shedding cycle makes the unobserved channels largely determined by the observed one.

b. Kolmogorov turbulence. The two-dimensional forced-turbulence vorticity dataset of Ref. 8, standard in the 2D generative-reconstruction literature: 40 independent trajectories of 320 frames at 256^2. We stride time by 4 (80 frames per trajectory, 3,200 total) and hold out entire trajectories: 32 for training, 8 for evaluation, assigned by a fixed seeded permutation. Cross-trajectory holdout is the 2D analogue of the spatial holdout below — reconstruction is scored on realizations of the flow the model has never seen at any time.

c. 3D isotropic turbulence. Four 125^3 cutouts at disjoint spatial locations of the JHTDB `isotropic1024coarse` DNS,[19] 50 snapshots each at a stride of 100 database steps, fields (U_x, U_y, U_z, p), per-field standardized. Training uses cutouts 1–3; all reported evaluation uses cutout 4, spatially disjoint from everything seen in training. Sensors observe U_x and U_z only; U_y and p are never observed, making this the identifiability probe of the suite.

C. Splits and leakage control

Every split in FlowRecon-Bench is chosen against the autocorrelation structure of its data, and the protocol is enforced by tooling rather than convention. Turbulence snapshots correlate at $r = 1.00$ frame-to-frame and $r = 0.67$ at separation 100; random-in-time splits therefore measure near-duplicate interpolation (Sec. V F quantifies the distortion). The suite uses spatial holdout (a disjoint cutout) for 3D turbulence, trajectory holdout for Kolmogorov, Reynolds-number holdout for the cylinder, and temporally blocked splits with decorrelation guard gaps wherever time enters. Model selection is separated from reporting: held-out snapshots are partitioned into a tuning half (odd indices) used for any checkpoint or hyperparameter selection and a test half (even indices) on which every reported number is frozen. The evaluation driver fingerprints the protocol — snapshot indices, sensor counts, and the index-sum of every seeded sensor draw — and aborts on mismatch, so no two methods can silently see different observations. Two incidents during construction illustrate why the tooling exists: an audit found one baseline's visual-evaluation path silently collapsing the third spatial axis, and the split machinery itself carried a one-frame boundary leak (an `int()` where floating-point arithmetic demanded rounding: at train fraction 0.8, $1 - 0.8$ evaluates below 0.2 and the first held-out-trajectory frame joined the training block). Both were caught by cross-checks the artifact now runs automatically.

D. Sensor protocols

The primary protocol draws sensors uniformly at random over the domain, per observed channel, at densities from 0.1% to 2% of the discretization (regime-dependent; 1% is the headline density), swept from 0.1% to 10% on Kolmogorov (Fig. 7), with counts and layouts redrawn per training step for amortized methods and frozen seeded draws at evaluation. Sensor layout is not incidental to difficulty: at fixed count, rearranging scattered sensors onto a uniform lattice changes reconstruction error by tens of percent (Sec. V F), so the layout is part of the protocol specification rather than an implementation detail. Two layout variants accompany the random-scatter default — a uniform lattice at matched count, and sensors confined to the cylinder wall ring (Sec. IV B) — and channel subsets are specified per regime (Sec. II B). Operator variants corrupt the assembled sensor sets in one place, after the draw and before any model sees them, by shared seeded code, so every method consumes bit-identical corrupted observations: Gaussian noise at $\sigma \in \{0.1, 0.3\}$ in standardized units, contiguous slab occlusion removing 25% of sensors, and channel dropout removing entire observed quantities (implemented, but not exercised in the reported results: the operator study runs on Kolmogorov, which observes a single channel; Table VI). Corrupted runs carry their own protocol stamp so they can never be aggregated with clean ones. Distinguishing per-frame, per-trajectory, and globally fixed sensor masks matters for comparability across papers, because the three make materially different problems out of the same nominal sensor budget. FlowRecon-Bench uses per-frame draws with frozen evaluation manifests, and says so explicitly, precisely so that a disagreement with a paper using another convention can be recognized as a protocol difference rather than a result.

E. Metrics

a. Accuracy, per channel. Relative L_2 per channel is primary, reported separately for observed and unobserved channels — aggregates average two different questions and are shown flagged, never headlined. For ensemble methods both the ensemble-mean error and the single-sample error are reported; the gap between them is the smoothing an ensemble average buys, and single-sample error measures whether individual realizations are physical.

b. Probabilistic scores. For an ensemble $\{s_k\}_{k=1}^{K}$ and truth y at a point, the fair CRPS estimator[14] is $\mathrm{CRPS} = \frac{1}{K}\sum_k |s_k - y| - \frac{1}{2K(K-1)}\sum_{k\neq l}|s_k - s_l|$, averaged over points and fields in standardized units. The spread–error ratio is the ratio of ensemble standard deviation (RMS over points) to the RMS error of the ensemble mean (ideal 1.0); $\mathrm{coverage}_{90}$ is the fraction of points whose truth falls in the central 90% ensemble interval (ideal 0.90); rank histograms accumulate the rank of the truth within the ordered ensemble. Deterministic methods enter the CRPS column exactly (CRPS reduces to MAE for a point forecast); their dispersion entries are undefined, not zero. Calibration metrics move with the operating point — sampler steps, ensemble size, sensor density — by more than they separate methods, so every calibration number carries its operating point and cross-method comparisons are made only at matched settings. Ensemble size deserves particular care: in a K-sweep on Kolmogorov (K=8–64, same frames and sensors), DMF-Gen's ensemble-mean rel-L_2 moved only 0.487→0.475 and CRPS was flat (the fair estimator is unbiased), but $\mathrm{coverage}_{90}$ rose 0.59→0.73 — finite-ensemble quantiles are systematically too narrow, so K=8 *understates* every generative method's calibration. We therefore report accuracy and CRPS at K=8 and treat K=8 coverage as a conservative bound, with the sweep in the artifact.

c. Physics metrics. Shell-averaged energy spectra of single samples (never ensemble means, which destroy small scales by construction), band ratios against the reference solution, structure functions, and velocity-gradient PDFs. One estimator detail is load-bearing enough to state in the protocol: non-periodic cutouts must be windowed before the FFT. On the 3D data the raw spectrum of the *truth* falls only 2.4× from $k = 32$ to the Nyquist wavenumber while a variance-preserving Hann taper restores a 5.8×10^5 decay; without the taper, a smooth Gaussian-process draw with zero energy at those wavenumbers registers the same broadband floor as the DNS, and spectral ratios computed there are ratios of two leakage floors that tend to unity for every method. All spectral numbers in FlowRecon-Bench use the windowed estimator.

d. Cost. Model selection is a cost–benefit decision, so cost is a first-class metric family, measured under the same harness for every method on every dataset: training seconds per optimizer step, total GPU-hours to the finished model, peak training memory, inference wall-clock per full-field reconstruction, and peak inference memory. Rows without cost entries are treated as incomplete by the table-assembly tooling. A separate resolution-scaling study (Sec. V E) measures memory versus output discretization from 64^3 to 1024^3, with out-of-memory walls reported as observed failures, not extrapolations.

F. Fairness methodology

Benchmark tables are only as credible as their weakest baseline, so the fleet is engineered adversarially against our own biases. (i) *Upstream fidelity audits*: every baseline implementation is diffed against its published code before its numbers are admitted, and every declared adaptation (dimensionality ports, set encoders, sampler retuning) is documented in the per-method audit notes; audit incidents (a conditioning defect in the latent flow-matching baseline, sampler-parameter sensitivity in S3GM) were repaired or labelled before reporting. (ii) *Parameter matching*: learned methods are matched to ∼6.5M parameters where the published configuration allows; declared waivers (CoNFiLD's published prior at 118.9M) are printed in the table. (iii) *Identical conditions*: the same data files, splits, augmentation, normalization statistics, seeded sensor draws, and metric estimator for every row. (iv) *Budget disclosure*: optimizer steps and wall-clock are reported as run, not silently equalized, and the table notes which budgets exceed which. We audited this rather than assuming it, comparing the configured against the as-run step count for all fourteen 2D rows. Every row ran to its configured epoch count, and all seven cylinder rows plus six of seven Kolmogorov rows land within 4% of ∼50k steps. The exception is Kolmogorov Geo-FNO at 200k steps, four times the rest, because its batch size is a quarter of the reference and the budget was fixed in epochs (Sec. IV C). The direction matters and we state it: the over-trained row is a *baseline that wins its regime*, not the benchmark's own method, so the error inflates a competitor rather than us — but it is still an error. The matched rerun has since replaced it (Sec. IV C): the ordering survives and the margin narrows from 21% to 14.5%. (v) *Tuning-budget disclosure*: the method we co-developed (DMF-Gen) received more configuration search than any baseline; upstream-faithful configurations are frozen as the headline rows and every tuned variant is a separately labelled arm. (vi) *Selection hygiene*: checkpoint selection uses the tuning split only (Sec. II C).

III. METHODS EVALUATED

The fleet spans four families; Table II records which methods can enter which regimes, itself a benchmark result, and Fig. 1 expands it to the per-method deployment axes — representation freedom, uncertainty, robustness, and scaling — that Sec. V measures. No method fills every column: DMF-Gen alone clears the representation axis but is the slowest sampler; latent FM inverts that profile exactly. In the present three-regime suite the "requires resampling" entries are exercised by exactly one dataset, the cylinder's body-fitted O-grid (Sec. II B), whose uniform-grid export exists only because grid-locked methods cannot consume the native mesh; Kolmogorov and the 3D turbulence cutouts are periodic Cartesian grids on which every family runs natively, so the interface filter is exercised here at its mildest, and the unstructured, case-varying geometries on which it becomes decisive are deferred to a follow-up study (Sec. VI). Per-method configurations, adaptations, and audit notes ship with the artifact.

a. Classical anchors. Four training-free floors run on every regime: nearest-neighbor interpolation (KD-tree), inverse-distance weighting over the $k = 8$ nearest sensors, gappy proper orthogonal decomposition[22] (POD basis from the training set, modal coefficients by least squares on the sensor readings, rank 80), and the training-mean constant. Distance handling is non-periodic by default — an earlier periodic implementation understated these floors on non-periodic cutouts and was corrected in audit — with periodic variants where the domain genuinely is periodic. As deterministic point forecasts their CRPS equals their MAE. These anchors define what any learned method must beat to justify its training cost.

b. Deterministic operator learners. Senseiver[5] is an attention-based point-native reconstructor mapping arbitrary sensor sets to query values through a latent bottleneck; it trains under the identical sensor randomization as the amortized generative methods. FNO3D places a spectral (FFT-grid) backbone inside the same conditional-generation framework as DMF-Gen, isolating the backbone axis from the formulation axis. DeepONet[23] enters in upstream-faithful form (branch/trunk with a declared set-encoder adaptation for variable sensor sets); its near-constant predictions are the pre-registered structural consequence of global mean pooling in the branch, and a structured-branch redesign ("DeepONet++", multiresolution binned pooling) is reported as a labelled improvement arm. Deterministic models return conditional-mean estimates: optimal in squared error, systematically smoothed, and structurally unable to report what the sensors leave undetermined.

c. Latent generative. Latent flow matching (LFM) is the latent-diffusion-family stand-in:[11,24] a convolutional autoencoder trained on full fields, followed by conditional rectified

FIG. 1. Per-method capability profile across the deployment axes measured in this paper (filled = yes, half-filled = partial or adapted, open = no). "Operator-robust" records behavior under the corrupted-operator variants of Sec. II D, whose full matrix belongs to the deferred multi-physics regime (Sec. VI); "dissipation-preserving" records the windowed dissipation-band ratios (Sec. V D); "memory flat vs resolution" and "sub-second inference" record the measured scaling study (Sec. V E); "recalibrable" records whether the post-hoc wrapper of Sec. V C applies. Axes are marked "no" only when the failure is structural.

flow in its latent space with a point-encoder conditioning branch. It is the strongest aggregate performer on 3D turbulence and carries the family's structural signature, a dissipation-range spectral floor inherited from the autoencoder.[25,26] CoNFiLD[11] is the published latent neural-field competitor: a conditional-neural-field autodecoder compressing each snapshot to a latent vector, an unconditional diffusion prior over latents, and diffusion posterior sampling for conditioning; three arms are reported (published prior; capacity-matched; 384-dimensional), with the latent-capacity ceiling of the autodecoder measured explicitly as the decoder bound.

d. Ambient generative. Gen4Turb[12] is voxel diffusion: an elucidated-diffusion 3D U-Net with mask-channel conditioning, run at its native 120^3 on the 3D data and retrained under the benchmark's observation model (the most generous variant is reported). S3GM[9] is a pretrained video-diffusion prior with sampling-time guidance, natively two-dimensional; its 3D entry is a declared adaptation treating one spatial axis as video time, and its published fixed guidance weights required tuning-split renormalization to converge at 3D scale. SiT-point adapts a scalable interpolant transformer[27] with a point tokenizer (not patchification): sensors and queries are tokenized directly, generation runs in fixed-size token chunks, and the chunk budget is the visible constraint. That is the 3D configuration; on the two 2D regimes the same implementation is run patch-tokenized, which is what those rows report, and on the cylinder it therefore reads the uniform-grid export rather than the mesh (Table II) (member-level chunk-incoherent speckle that ensemble averaging hides — one reason FlowRecon-Bench reports member-level metrics). DMF-Gen[28] performs conditional

TABLE II. Capability matrix: which method families can honestly enter which regimes. RS = requires resampling to a grid (lossy near the body-fitted boundary of the cylinder mesh); in this suite only the cylinder exercises that column. §SiT is point-tokenized in the 3D run but patch-tokenized in both 2D runs, so on the cylinder it consumed the uniform-grid export rather than the mesh: the family can enter grid-free, this particular configuration did not. Every cell in the cylinder column — the only column that exercises the resampling distinction — was checked against the data export the corresponding run actually opened, not against the family's nominal capability: Senseiver, MLP-RBF and DMF-Gen read the 23,800-cell mesh, while Geo-FNO, latent FM, S3GM and SiT read the 400×200 grid export. We keep families rather than one row per method, because the claim the table makes is about interfaces, and methods inside a family share the interface even when their accuracy differs by a factor of three.

Family	Cyl.	Kolm.	JHU
Interpolation/POD anchors	✓	✓	✓
Point-attention det. (Senseiver)	✓	✓	✓
Grid det. (FNO-class)	RS	✓	✓
DeepONet-class	✓	✓	✓
Latent-grid generative (LFM)	RS	✓	✓
Latent neural-field gen. (CoNFiLD)	✓	✓	✓
Voxel/video diffusion (Gen4Turb, S3GM)	RS	✓	✓
Point-token generative (SiT-point)	RS§	✓§	✓
Ambient point-cloud FM (DMF-Gen)	✓	✓	✓

rectified flow directly on field values at scattered points — ambient function space, no autoencoder, no voxelization — with a Gaussian-process source shared across discretizations, conditioning factored into a global cross-attention summary plus a local k-nearest-sensor retrieval per query, a Monte-Carlo subsampled training objective (a random $\sim$2% of points per step), and training amortized over sensor counts, layouts, noise, and observed-channel subsets. We evaluate it as published, at the benchmark's parameter budget; it is one row of the fleet, not the paper's contribution.

IV. RESULTS BY REGIME

A. Laminar vortex shedding: where classical methods should suffice

The dataset was generated for this benchmark (Sec. II B) and verified against the literature before any model saw it: Strouhal numbers 0.140/0.160/0.173/0.187/0.200/0.207 for Re 60–250 track the Williamson–Norberg correlations, mean drag sits where the 6% blockage predicts, and lift amplitude grows monotonically with Re. Sensors observe (u, v) at 1% of the 23,800 mesh cells (238 per field); pressure is never observed; evaluation is on the held-out Reynolds numbers $\{80, 250\}$.

The classical floors decide this regime before the learned fleet reports. The training family's POD spectrum is essentially exhausted at rank 20 (99.7% energy), and *gappy POD with 20 modes reconstructs held-out Reynolds numbers at 0.056 relative* L_2 *— including the never-observed pressure at 0.05* — because the modal coefficients inferred from sparse velocities carry the full coupled state. Pure interpolation cannot do this (p column: 1.0 by construction) and trails badly on the velocities as well (IDW 0.71 aggregate). This is the mirror image of 3D turbulence, where the same POD basis was useless (0.95, Table VIII) and interpolation was the strong floor: *which classical method is the one to beat is itself regime-dependent*, and a learned method only earns its complexity here if it approaches the POD floor at comparable cost.

It does not, and the gap is resolved rather than marginal: on the same 50 frames and the same seeded draws, rank-20 gappy POD beats the best learned method by 0.037 relative L_2 (95% bootstrap CI $[-0.042, -0.032]$, Wilcoxon $p < 10^{-14}$). The best learned method here is Geo-FNO, which reads the grid export and so, by protocol, receives its sensors on the grid rather than the mesh; that pairing is over frames, not sensor positions. Against the mesh-native methods, which share the classical floors' mesh, the comparison is paired in both: re-scored on the floors' exact sensor draw, the three mesh rows move by at most 0.006 (Senseiver 0.140, MLP-RBF 0.394, DMF-Gen 0.221; the table reports the original runs) and gappy POD's lead stays resolved against each (-0.084, -0.339, -0.165; all $p < 10^{-14}$). Table V reports the fleet: the closest learned method, Geo-FNO at 0.093, is still 1.7× the rank-20 POD floor, and the generative models trail further (latent FM 0.192, DMF-Gen 0.221, SiT 0.328). Between learned rows, which share one sensor draw and are fully paired, DMF-Gen places fourth: behind latent FM by 0.030 (95% CI $[0.014, 0.044]$) and behind Geo-FNO by 0.128 ($[0.123, 0.134]$, better on 0 of 50 snapshots). The ordering on the never-observed pressure channel is the sharper statement — gappy POD recovers p at 0.043–0.051, better than any learned method (Geo-FNO 0.088, Senseiver 0.095, DMF-Gen 0.128) — because a basis fitted to the coupled state propagates sparse velocity observations into pressure, whereas the learned models must infer that coupling from data alone. Read against Sec. V B, where *no* method recovers an unobserved channel of 3D turbulence, this is the benchmark's cleanest statement that cross-channel identifiability is a property of the regime and not of the method class.

Two rows carry caveats rather than results. S3GM diverges here (1.019, worse than predicting the training mean) under its published predictor–corrector guidance at this benchmark's parameter budget; we report it as a documented sampler failure with its diagnostic figures in the artifact rather than dropping the row, since a benchmark that hides the arms that break is not measuring robustness. Latent FM is the one over-dispersed model in the suite (spread–error 1.26, 90% coverage 0.89 against a 0.90 target), which is what a calibrated-but-blurry posterior looks like and is consistent with its strong CRPS at moderate relative L_2. DMF-Gen sits on the other side of unity (spread–error 0.57, coverage$_{90}$ 0.51), so the two generative routes miss calibration in opposite directions on the same flow — an argument for the post-hoc recalibration of Sec. V C rather than for either architecture.

Which held-out Reynolds number? The cylinder's held-out set is two Reynolds numbers, and averaging them hides the more informative question. The training set is $Re \in \{60, 100, 150, 200\}$, so $Re = 80$ is an *interpolation* in Reynolds number — bracketed by two training values — while $Re = 250$ lies past the training maximum and is a genuine *extrapolation*. Splitting the same 50 canonical frames along that line (Table III) gives three results.

First, the ranking does not change: gappy POD leads at both Reynolds numbers, Geo-FNO is second at both, Senseiver third at both. The ordering this regime reports is not an artifact of averaging a hard case with an easy one.

Second, extrapolating costs every working method between 1.26× and 1.54×, and the cost is largest for the *most accurate* rows — gappy POD 1.54× (0.044 to 0.068), Geo-FNO 1.50× — and smallest for the rows already sitting near the training-mean floor (interpolation 1.11×; diverged S3GM 1.01×, which cannot get worse). Methods that exploit the training family's structure hardest are the ones that lose most when asked to leave it, which is the honest counterweight to the headline that classical methods win this regime: gappy POD wins it because the family is low-rank, and it gives back a larger share of its margin than anything else in the table the moment the family is left behind.

Third, the never-observed pressure channel is the most Reynolds-sensitive part of the reconstruction. Six of the seven learned and classical rows that recover p at all degrade *faster* on p than on their aggregate (gappy POD 1.73× vs 1.54×; Senseiver 1.66× vs 1.26×; DMF-Gen 1.51× vs 1.30×; SiT is the exception at 1.13× vs 1.31×). This is what should be expected and is worth stating because it is rarely measured: an unobserved channel is reconstructed entirely from cross-channel structure inferred at the training Reynolds numbers, so it is the first thing to fail when the flow leaves them. It also explains the scalar-recalibration failure reported in Sec. V C — a single multiplier fitted across a held-

TABLE III. Cylinder cross-Reynolds breakdown of Table V, over the same 50 canonical frames (25 per Reynolds number). $Re = 80$ is an interpolation in Re; $Re = 250$ is an extrapolation past the training maximum of 200. Aggregate relative L_2, their ratio, and the never-observed pressure channel. Interpolation rows return the training mean on p by construction, and their aggregates coincide to four decimals by coincidence, not by error — both are dominated by that p=1.0 column.

	aggregate			p (never observed)	
Method	*Re* 80	*Re* 250	ratio	*Re* 80	*Re* 250
DMF-Gen	0.192	0.250	1.30	0.102	0.154
SiT	0.284	0.373	1.31	0.258	0.291
Latent FM	0.168	0.215	1.28	0.143	0.188
S3GM (PC-DPS)	1.013	1.025	1.01	1.007	1.002
Senseiver	0.129	0.162	1.26	0.071	0.118
MLP-RBF	0.325	0.452	1.39	0.177	0.265
Geo-FNO	0.074	0.111	1.50	0.069	0.107
Gappy POD (r=20)	0.044	0.068	1.54	0.037	0.064
IDW (k=8)	0.667	0.742	1.11	1.000	1.000
Nearest neighbour	0.667	0.742	1.11	1.000	1.000
Training mean	1.000	1.000	1.00	1.000	1.000

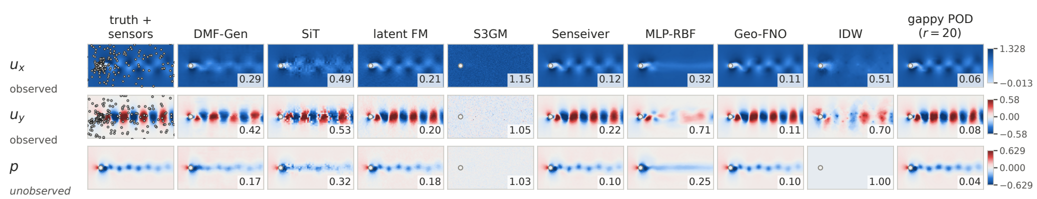

FIG. 2. Cylinder wake, held-out Re = 250 frame (val 300). Rows are the three reconstructed channels: u_x and u_y observed at 238 sensors per field (1%; circles on the truth panels) and p never observed. Columns are the truth and then one baseline each. Mesh-native methods (DMF-Gen, Senseiver, MLP-RBF, IDW, gappy POD) share one sensor draw; grid-native methods (SiT, latent FM, S3GM, Geo-FNO) receive the same count on the grid export, exactly as in Table V. Generative columns show one posterior sample, not the ensemble mean; the number in each panel is that field's single-sample relative L_2 in standardized units, so it sits above the table's ensemble-mean error. The p row is the point: gappy POD infers the unobserved channel through the basis's cross-channel correlations, S3GM's sampler diverges on this dataset, and interpolation returns the training mean by construction.

out set spanning an interpolation and an extrapolation is fitting two different error scales at once.

B. The same flow, sensed from the surface

Every result above places sensors throughout the volume. Experiments usually cannot: a wind-tunnel model carries pressure taps on its skin, and the flow away from the body is exactly what is unmeasured. We therefore repeat the cylinder task with sensors confined to the 360 wall-adjacent cells, all three fields read at the taps and nothing observed off the body, keeping the flow, the held-out Reynolds numbers, the frames, the seeded-draw machinery and the training budget identical. Predictions were registered before the runs and are reported whichever way they fell; two of the three failed.

The classical floor does not survive the change of geometry. Gappy POD with 20 modes wins the volume task outright at 0.056 (Table V) and reaches only 0.689 from the full tap

TABLE IV. Cylinder *surface-to-field*: reconstruct the full wake from taps on the body alone, held-out Re $\in \{80, 250\}$, 50 Re-stratified frames. Columns are taps per field on the 360-cell wall ring; all three fields are read at the taps and nothing is observed off the body. † = identical to the 64-tap entry by construction: the ring rasterizes onto 62 grid cells, so grid-locked methods cannot represent a larger budget. Compare with Table V, the same flow sensed in the volume, where the ranking is inverted.

	Rel. L_2 at n taps/field				
Method	32	64	128	360	CRPS
Senseiver	0.109	0.105	0.104	**0.103**	0.058
Geo-FNO	0.161	0.119	0.119	0.119	0.073
DMF-Gen	0.271	0.252	0.247	0.245	0.111
MLP-RBF	0.337	0.307	0.269	0.246	0.138
SiT	0.494	0.476	0.476	0.476	0.180
Latent FM	0.631	0.569	0.569	0.569	0.220
S3GM	0.897	0.891	0.891†	0.891†	0.468
Gappy POD (r=20)	1.223	0.919	0.781	0.689	0.306
IDW (k=8)	2.077	2.117	2.132	2.141	1.937
Nearest neighbour	2.159	2.153	2.148	2.144	1.940

ring, while Senseiver improves from 0.145 to 0.103 and now leads by a factor of 6.7 over the classical anchor (Table IV). We had pre-registered the opposite risk — that POD might win here too, since the wake is genuinely low-rank — and it did not. The reason is that a modal basis needs its coefficients to be *identifiable* from the measurements, and a boundary ring samples a one-dimensional manifold on which the leading modes are nearly degenerate; low-rank structure in the field does not imply well-conditioned inference from an arbitrary sensor set. Interpolation fails completely, at 2.1 relative L_2, worse than predicting the training mean, because it cannot extrapolate off the sensor manifold at all.

The rasterization cost is exact, not approximate. The 360-cell ring maps onto only 62 distinct cells of the 400×200 export, so every grid-locked method is blind to tap budgets above 62: Geo-FNO, SiT and latent FM return bit-identical errors at 64, 128 and 360 taps, to six decimal places, while the three mesh-native methods keep improving to 360. S3GM is grid-locked by the same argument and its 128- and 360-tap entries are marked † because they are *inferred* from the pool cap rather than re-measured — we spent the sampler budget on densities that could differ instead. The identity is a measurement for three of the four rows and a construction for the fourth, and the table distinguishes them. This is the capability matrix's "requires resampling" column measured rather than asserted, and it is the clearest evidence in the paper that the interface a method demands is a scientific property and not an implementation detail.

What the task does not show. Grid-locked is not the discriminator: Geo-FNO places second at 0.119, ahead of every point-native generative model. The surface geometry separates deterministic from generative methods, the reverse of what we expected from the argument that an under-determined far-wake posterior should reward a calibrated distribution. DMF-Gen is fourth at 0.245 and does not lead on CRPS either. We record the failed prediction rather than reframing it.

C. Two-dimensional Kolmogorov turbulence

Sensors observe vorticity at 1% of the 256^2 points (655 sensors); learned models are scored on 50 evenly spaced frames of the eight held-out trajectories (K=8, identical canonical seeded draws across methods; classical floors on all 640). Table VII reports the full fleet alongside the floors, and Fig. 7 (left) the sensor-count sweep. Three regime findings stand

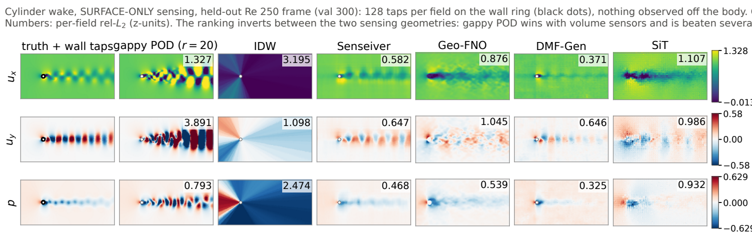

FIG. 3. Surface-only sensing on the held-out Re = 250 frame: taps (black) sit on the body, everything else is inferred. Read against Fig. 2, the identical frame sensed in the volume.

TABLE V. Cylinder wake, cross-Re evaluation on the held-out Re $\in \{80, 250\}$ (u, v observed at 1% = 238 sensors per field on the 23,800-cell mesh; *p is never observed*; learned rows: 50 Re-stratified held-out frames, K=8; classical floors: all 600). Deterministic rows carry K=1 and null dispersion. Interpolation returns the training mean on the unobserved channel, hence p=1.000 by construction.

	Rel. L_2						
Method	agg.	u_x	u_y	p	CRPS	Cov.$_{90}$	s/field
DMF-Gen (NFE 4)	0.221	0.182	0.353	0.128	0.098	0.51	0.08
SiT (50 steps)	0.328	0.302	0.409	0.274	0.106	0.55	0.19
Latent FM (NFE 4)	0.192	0.170	0.239	0.166	0.076	0.89	0.03
S3GM (PC-DPS)	1.019	1.035	1.017	1.005	0.521	0.58	29.24
Senseiver	0.145	0.101	0.240	0.095	0.080	—	0.01
MLP-RBF	0.388	0.265	0.679	0.221	0.213	—	0.01
Geo-FNO	0.093	0.087	0.103	0.088	0.053	—	0.01
Gappy POD (r=20)	**0.056**	0.043	0.075	0.051	0.029	—	—
Gappy POD (r=80)	0.067	0.055	0.103	**0.043**	0.028	—	—
IDW (k=8)	0.707	0.504	0.616	1.000	0.440	—	—
Nearest neighbour	0.707	0.508	0.613	1.000	0.421	—	—
Training mean	1.000	1.000	1.000	1.000	0.664	—	—

out.

First, *the 3D ranking does not transfer*. Geo-FNO wins relative L_2 outright at 0.417 — 14.5% better than the best generative model — while latent FM, the 3D aggregate winner, trails even the interpolation floor here (0.599 against 0.531) at the matched ∼50k-step budget, and DMF-Gen beats it on both relative L_2 and CRPS on all 50 paired snapshots. The remaining generative models land between them (DMF-Gen 0.487, SiT 0.494, S3GM 0.498); of these only DMF-Gen is ambient point-based here, since SiT and S3GM are run patch- and voxel-tokenized on this grid. The two deterministic point models split: MLP-RBF at 0.639 and Senseiver at 0.913, the latter barely inside the training-mean floor.

Second, *the accuracy and the probabilistic ranking disagree across method families, not within them*. The same Geo-FNO row that wins relative L_2 is beaten on CRPS by three of the four generative models (0.287 against S3GM 0.231, SiT 0.234, DMF-Gen 0.235; only latent FM, at 0.310, is worse), and the split is resolved on paired snapshots: Geo-FNO trails DMF-Gen on CRPS by 0.052 (95% CI $[0.048, 0.057]$, all 50), while DMF-Gen and SiT are indistinguishable on it (0.001, $[-0.002, 0.004]$). This is the distortion–perception tradeoff of Sec. V A appearing as a family-level split rather than a curve: a conditional-mean estimator minimizes squared error and is then penalized by a proper score that rewards a calibrated distribution. Reporting either metric alone would name a different winner, which is why the benchmark requires both.

The winning row initially received more training than the rest, and we retrained it to check. The Kolmogorov budget was set in *epochs* (2500) at batch size 128, giving the

$\sim$50k optimizer steps every other row receives: Senseiver 50.0k, MLP-RBF 50.0k, latent FM 50.0k, DMF-Gen 51.2k, S3GM 51.2k, SiT 52.0k. Geo-FNO inherited the same epoch count at batch size 32, which is 80 rather than 20 steps per epoch and therefore 200k steps, four times the fleet budget. Matching budgets by epoch count rather than by step count is an easy error to make when batch sizes differ across a fleet, and it is why this benchmark reports as-run steps per row.

Retrained at exactly 50k steps (625 epochs, configuration otherwise untouched), *Geo-FNO still wins*: 0.417 relative L_2 against DMF-Gen's 0.487, SiT's 0.494 and S3GM's 0.498. The margin narrows from 21% to 14.5% and the ordering is unchanged, with the paired difference resolved against all three (-0.071, CI $[-0.079, -0.063]$ versus DMF-Gen; $p < 10^{-14}$ in every case) and roughly seventeen times the training-seed spread. The four-fold budget was worth 0.032 relative L_2 to this architecture — a real effect, and smaller than the gap it was suspected of manufacturing. We report the matched number as the row and the over-trained one as an ablation, because the leaderboard should answer what a method achieves at the benchmark's budget.

Third, the floors carry regime information of their own: gappy POD's rank-80 training basis captures only 70.5% of the training-set energy — two-dimensional turbulence is not low-rank in the POD sense, unlike the cylinder — and plain inverse-distance weighting beats it by 20% relative L_2.

All comparisons here are paired: every method is scored on the same 50 frames under the same seeded draws, so the per-snapshot difference is the unit of analysis. We report the mean paired difference with a 95% percentile bootstrap interval (20,000 resamples) and call a gap resolved only when the interval excludes zero *and* a Wilcoxon signed-rank test agrees at $p < 0.05$. Geo-FNO's lead is resolved against the whole fleet (against DMF-Gen, -0.071, CI $[-0.079, -0.063]$, $p < 10^{-14}$), whereas the three generative models behind it are not separable from each other: DMF-Gen versus SiT gives -0.006 with $p = 0.16$, and SiT versus S3GM -0.004 with $p = 0.32$. Only the DMF-Gen–S3GM pair separates (-0.010, $p = 0.008$), which is the expected intransitivity when three methods sit inside a 0.011 band — we therefore report them as a tie group rather than ranking them.

Retraining answers the same question from the other side. Each 2D method was retrained from scratch under two further seeds (7 and 1337, configurations otherwise identical) and re-evaluated under the frozen protocol. Training-seed spread is small: pooled standard deviation 0.0041 relative L_2 across the full Kolmogorov fleet and 0.0081 on the cylinder. The most seed-sensitive row is S3GM in both regimes (0.0082 and 0.0163), which is consistent with the sampler fragility documented elsewhere; the least is SiT on Kolmogorov (0.0006). Every adjacent gap in the cylinder ranking exceeds the pooled spread by at least a factor of 3.7 — the tightest being latent FM against DMF-Gen at 0.030 — so that ordering is seed-robust end to end, though by a smaller margin than the accuracy gaps alone suggest. On Kolmogorov the same test isolates the same tie group the paired intervals found: Geo-FNO's lead over DMF-Gen is 0.071, seventeen times the seed spread (Geo-FNO's own replicates were trained at its original 4$\times$ budget, so its seed spread is measured there), while DMF-Gen–SiT (0.006) and SiT–S3GM (0.004) are one to one-and-a-half seed standard deviations apart and cannot be called ordered. Two independent notions of uncertainty — resampling the evaluation frames and resampling the training run — agree on which comparisons this benchmark resolves, which is the reassurance a leaderboard needs before anyone builds on its ordering.

The measurement operators of Sec. II D put the same ordering under corrupted observations (Table VI). Gaussian noise at $\sigma = 0.1$ is invisible at this precision: no row moves by more than 0.007. At $\sigma = 0.3$ the four leading rows lose 6–12% (Geo-FNO 0.417 $\to$ 0.457, DMF-Gen 0.487 $\to$ 0.525, SiT 0.494 $\to$ 0.555, S3GM 0.498 $\to$ 0.526). A contiguous slab removing a quarter of the sensors costs those rows far more, 31–44%, because it leaves a region with no nearby data rather than thinning the data everywhere. Geo-FNO keeps its lead under every operator (0.554 occluded, against 0.651 for the next row). Below it the clean tie group reshuffles: SiT degrades most (1.44$\times$) and ends level with latent FM (0.709 against 0.695), which it led by 0.105 on clean data, while S3GM (0.651) and DMF-Gen

TABLE VI. Measurement operators on Kolmogorov at 1% sensing (655 sensors): ensemble-mean relative L_2 over the 50 canonical test frames (K=8 for generative rows, one forward pass for deterministic ones). Every method receives bit-identical corrupted observations (Sec. II D); σ is Gaussian noise in standardized units, and the slab removes one contiguous region holding 25% of the sensors. Last column: occluded over clean. Geo-FNO is the budget-matched run.

Method	clean	σ=0.1	σ=0.3	25% slab	slab/clean
Geo-FNO	0.417	0.421	0.457	0.554	1.33
DMF-Gen	0.487	0.492	0.525	0.660	1.35
SiT	0.494	0.501	0.555	0.709	1.44
S3GM (PC-DPS)	0.498	0.503	0.526	0.651	1.31
Latent FM	0.599	0.601	0.608	0.695	1.16
MLP-RBF	0.639	0.640	0.645	0.763	1.19
Senseiver	0.913	0.913	0.915	0.920	1.01

(0.660) stay 0.009 apart — about two clean-fleet seed standard deviations — and we do not rank them. The rows that barely notice any operator are the ones with the highest clean error: Senseiver moves from 0.913 to 0.920 under occlusion, and latent FM and MLP-RBF lose 16–19%. Read alone, the last column of Table VI would rank the least accurate model as the most robust; insensitivity to losing sensors is only a virtue in a model that was using them. Channel dropout is not in the table: Kolmogorov observes a single channel, so removing it would leave nothing to condition on.

Spectrally the fleet separates differently again (Fig. 4). Measured against the truth spectrum of the same frame over a small-scale band (k = 32–128), the samplers that inject texture overshoot — latent FM carries 1.17 of the true energy, SiT 1.72 and S3GM 1.87 — while DMF-Gen carries 0.139 and the deterministic rows almost none (Geo-FNO 0.048, Senseiver 0.068, MLP-RBF 0.043, gappy POD 0.021). The ordering is the reverse of the 3D result, where the latent decoder truncated the smallest scales and the ambient point model retained them (Sec. V D). Ratios above unity are errors too: small-scale energy in excess of the truth is spurious texture rather than resolved structure, and the nearest-neighbour interpolant reaches 2.55 in the same band through piecewise-constant blocking. The accuracy leader carries almost none of that band — Geo-FNO wins relative L_2 with 0.048 of the true small-scale energy — so the distortion–perception split of Sec. V A reappears here as a spectral one. The samplers ran at their own budgets (DMF-Gen and latent FM at NFE 4, SiT at 50 steps, S3GM at 200), and a coarse solver budget can bias a sample smooth, but that is not the explanation here: re-sampling DMF-Gen on the same frame, sensors and seed at SiT's 50 steps raises its small-scale energy only from 0.139 to 0.185 of the truth (0.174 at 16 steps), against SiT's 1.72 at the same budget, while its large-scale excess grows (1.40 to 1.67) and its single-sample error worsens (0.71 to 0.78). The 2D small-scale deficit is architectural, not a coarse-solver artifact.

D. Three-dimensional isotropic turbulence: the full fleet

The 3D turbulence benchmark is the controlled comparison: a periodic-topology regular grid is native terrain for voxel, patchified, and latent-grid methods, so every family competes at full strength. Sensors observe U_x and U_z at 1% of 1.95×10^6 points per channel; U_y and p are never observed. Table VIII reports per-channel accuracy and probabilistic scores for the full fleet on all 50 held-out snapshots of the spatially disjoint cutout under canonical seeded draws; Table IX reports the matched cost accounting.

The honest headline is that no method leads every column. SiT-point posts the best observed-channel errors (0.143/0.132) at roughly 150 s per field — 239 sequential 8,192-

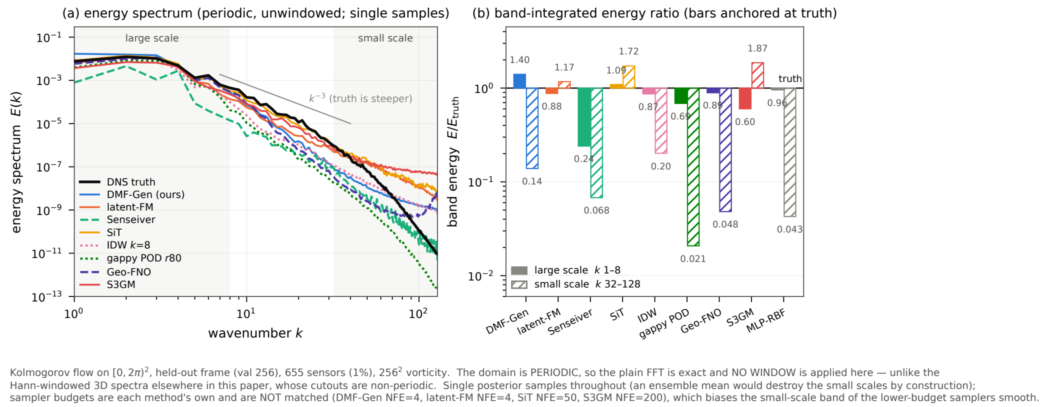

FIG. 4. Kolmogorov spectral fidelity at 655 sensors (1%), held-out frame, one reconstruction per method. (a) Shell-averaged energy spectra $E(k) = \sum_{\mathrm{shell}} |\hat{\omega}(k)|^2/2k^2$ on integer shells to the Nyquist limit. The field occupies the doubly periodic box $[0, 2\pi)^2$ and the saved frame is a full period, so no window is applied here, unlike the Hann-windowed 3D spectra of Fig. 8, whose cutouts are non-periodic sub-blocks; the estimators differ because the boundary conditions differ. The k^{-3} line is a reference slope, not a fit: the DNS is steeper at this Reynolds number. (b) Band-integrated energy relative to truth over $k = 1$–8 and $k = 32$–128, on a log axis anchored at unity so deficit and spurious excess both read as error. All generative entries are single posterior samples; MLP-RBF, a ninth series past the validated eight-colour palette, appears in (b) only. Sampler budgets are each method's own; the NFE-matched control in the text shows the budget does not explain the ordering.

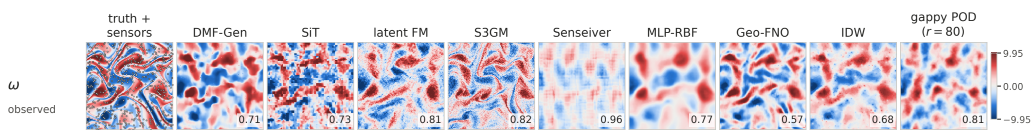

FIG. 5. Kolmogorov 256^2, canonical held-out frame (val 256). Vorticity is the only channel and is observed at 655 sensors (1%; dots on the truth panel), with the identical draw in every column. Columns and per-panel numbers as in Fig. 2. Gappy POD is visibly over-smoothed — two-dimensional turbulence is not low-rank — the generative samples carry small-scale texture the deterministic rows lack, and the patch tokenizer's block structure is directly visible in the SiT panel.

token chunks at 32 steps — and its single samples carry chunk-incoherent speckle that the ensemble mean averages away. Latent flow matching leads the aggregate, the unobserved channels, and CRPS: its convolutional decoder emits globally structured fields even where no sensor constrains them, at the dissipation-range price quantified in Sec. V D. Non-periodic inverse-distance weighting — a training-free interpolant — edges the best learned observed-channel error (0.174 vs. 0.176 on U_x for DMF-Gen), so no learned method beats interpolation on observed channels at 1% density; the learned and generative payoffs concentrate where the problem is ill-posed — unobserved channels, sparser sensing — and in sampling cost (DMF-Gen reaches its accuracy in 4 network evaluations against 32–1000 for the guided-diffusion and transformer baselines). DeepONet's near-constant predictions are the structural consequence of global mean pooling in its branch; the labelled DeepONet++ arm (structured pooling) improves U_x from 0.593 to 0.512 and remains behind the point-attention family.

S3GM needs its three rows read together, because reporting only the first would misrepresent the method. At the published guidance coefficients (α_{case}=0.5, β=0.4) the sampler diverges at this scale by five orders of magnitude, and we verified that each coefficient di-

TABLE VII. Kolmogorov 256^2, trajectory-holdout evaluation (vorticity observed at 1% = 655 sensors, identical canonical seeded draws; learned rows: 50 evenly spaced held-out frames, K=8; classical floors: all 640). Deterministic rows carry K=1 and null dispersion, so their coverage cell is empty, their CRPS equals the MAE exactly, and their mean and single-sample columns coincide. Inference cost per field on one H100. Rows are generated from the canonical JSONs by `src/figs_pof/make_2d_tables.py`.

	Rel. L_2					
Method	mean	sample	CRPS	Cov.$_{90}$	s/field	GB
DMF-Gen (NFE 4)	0.487	0.578	0.235	0.59	0.28	0.76
SiT (50 steps)	0.494	0.590	0.234	0.64	0.18	0.24
Latent FM (NFE 4)	0.599	0.731	0.310	0.64	0.03	0.19
S3GM (PC-DPS)	0.498	0.700	**0.231**	0.80	22.05	17.42
Senseiver	0.913	0.913	0.697	—	0.01	0.71
MLP-RBF	0.639	0.639	0.469	—	0.01	0.79
Geo-FNO	**0.417**	**0.417**	0.287	—	0.01	0.34
IDW (k=8) (per.)	0.531	0.531	0.375	—	—	—
IDW (k=8) (non-per.)	0.546	0.546	0.385	—	—	—
Nearest neighbour (per.)	0.582	0.582	0.371	—	—	—
Nearest neighbour (non-per.)	0.588	0.588	0.375	—	—	—
Gappy POD (r=80)	0.680	0.680	0.510	—	—	—
Training mean	1.000	1.000	0.753	—	—	—

verges independently — this is a property of the guidance at 1.95M points, not a porting defect, and the archived divergence figures are documentation of the failure rather than a model result. Reducing both coefficients by two orders of magnitude makes the sampler stable and it then scores 0.682 aggregate; replacing the guidance with a normalized form whose coefficients are fitted on the tuning split reaches 0.628 with observed-channel errors of 0.260 and 0.254. Both are labelled deviations from the published method, so neither can be quoted as "S3GM" without qualification, and neither closes the unobserved-channel gap (0.98–1.02) that no method in the fleet closes. The benchmark's rule — evaluate as published, then report labelled repairs separately — is what makes the distinction visible instead of a footnote.

The top of this table is partly a tie. Applying the paired analysis of Sec. IV C to the canonical n=50 JSONs, latent FM's lead is resolved against both of the rows immediately behind it (against FNO3D -0.118, CI $[-0.147, -0.088]$; against DMF-Gen -0.124, CI $[-0.147, -0.102]$; both $p < 10^{-9}$), while FNO3D and DMF-Gen are not separable from each other (-0.007, CI $[-0.027, +0.015]$, $p = 0.33$) despite being reported as different rows. A ranking that reads the third and fourth entries as ordered would be over-reading the benchmark's resolution at this n.

Two limits on that statement are worth being explicit about, because both bound what this table can support. First, only the three rows above could be compared frame by frame: the SiT-point, Senseiver, Gen4Turb and DeepONet rows have no canonical per-snapshot payloads in the artifact set (SiT and Gen4Turb have none; Senseiver's predate the current schema), and a paired test cannot be reconstructed from an aggregate. Their positions in Table VIII are therefore point estimates without a paired interval, and we do not claim any of them is separated from a neighbour. Second, the seed error bar reported in this paper covers the two 2D regimes only (Sec. IV C); the 3D rows are single training runs, so the resolution argument above rests on evaluation-frame resampling alone rather than on both notions of uncertainty. Where a 3D comparison is called resolved it is resolved in that weaker sense, and we flag it rather than let the 2D seed study stand in for a measurement we did not make at 3D scale.

TABLE VIII. 3D isotropic turbulence, spatially disjoint evaluation region (1.95M points; U_x, U_z observed at 1% per channel, U_y^*, p^* unobserved; canonical seeded sensor draws, all n=50 held-out snapshots, K=8 ensembles at each method's listed sampler setting). Per-channel relative L_2 of the ensemble mean is primary; agg† averages all four channels and is flagged, not headlined. Unobserved-channel error ≥ 1.0 is worse than predicting the training mean; values near 1.0 across the fleet reflect a shared identifiability limit of point-conditioned methods (Sec. V B). Spread–error and coverage ideals are 1.0 and 0.90; for deterministic methods CRPS reduces to MAE and dispersion is undefined. Step budgets are as-run, not equalized (Table IX). Labelled improvement arms (Senseiver capacity, DeepONet++, CoNFiLD repairs, S3GM normalized guidance) are reported in the artifact. ‡ = guidance deviates from the published setting, which is why the upstream-faithful S3GM row is reported separately above it rather than replaced by the working one; the normalized-guidance coefficients were fitted on the tuning split only, and its tune/test difference is below 0.001 relative L_2. A CoNFiLD strict-2048 arm was run and is deliberately *not* in this table. Its 50 per-snapshot files aggregate to rel-L_2 0.941 / CRPS 0.500 (U_x 0.525, U_z 0.559) rather than the 0.90 recorded for it earlier, and they carry no protocol stamp — no sensor count, conditioned channels, seed or K — so there is no way to confirm from the artifact that it saw the canonical sensor draw. We report it here in the text rather than as a row, because admitting a number to a comparison table on the strength of its filename is exactly the failure this benchmark exists to argue against.

Method	U_x	U_y^*	U_z	p^*	agg†	CRPS	Spr/Err	Cov.$_{90}$
DMF-Gen (NFE 4)	0.176	1.048	0.182	0.964	0.593	0.291	0.63	0.54
Latent flow matching	0.248	**0.727**	0.257	**0.646**	**0.469**	**0.244**	0.68	0.56
SiT-point (N=32)	**0.143**	0.968	**0.132**	0.937	0.545	0.338	0.75	0.58
FNO3D (same framework, NFE 4)	0.188	0.904	0.191	1.061	0.586	0.269	**0.94**	**0.63**
CoNFiLD (published prior, DPS 1000)	0.409	0.751	0.374	1.007	0.635	0.371	0.82	0.57
CoNFiLD (capacity-matched)	0.488	1.089	0.420	1.487	0.871	0.562	0.62	0.45
CoNFiLD (384-d)	0.524	0.604	0.451	1.401	0.745	0.480	0.69	0.50
Gen4Turb (anneal, 32-step)	0.224	0.996	0.455	1.005	0.670	0.407	0.37	0.33
S3GM (upstream guidance, N=200)	diverges at 3D scale (rel-L_2 8.9×10^5); see text							
S3GM (retuned guidance)‡	0.362	0.992	0.350	1.024	0.682	0.368	0.48	0.42
S3GM (normalized guidance)‡	0.260	0.980	0.254	1.018	0.628	0.343	0.52	0.46
Senseiver (deterministic)	0.361	1.077	0.380	0.978	0.699	0.479	—	—
DeepONet (deterministic)	0.593	0.972	0.652	0.985	0.800	0.570	—	—
Nearest neighbor (KD-tree)	0.209	1.000	0.219	1.000	0.607	0.406	—	—
IDW (k=8, non-periodic)	0.174	1.000	0.181	1.000	0.589	0.396	—	—
Gappy POD (r=80)	0.523	1.306	0.540	1.443	0.953	0.673	—	—
Training mean	1.000	1.000	1.000	1.000	1.000	0.786	—	—

V. CROSS-CUTTING ANALYSES

A. The distortion–perception tradeoff, operationally

Relative L_2 of the ensemble mean is minimized by the posterior mean — by construction it rewards collapse toward a smooth conditional average — while proper scores and physics metrics reward calibrated diversity and sharp members. The benchmark measures the tension directly through the sampler's operating point: for the rectified-flow methods, two integration steps give the best mean-field error and the sharpest single samples, four steps are the CRPS optimum, and dispersion rises monotonically with further steps at flat CRPS (spread–error 0.65 at NFE 2 to 0.92 at NFE 16 for the λ_{sp}=0 DMF-Gen variant). A spectral training penalty moves along the same frontier: it buys single-sample sharpness and inertial-band fidelity at a measured cost in dispersion. The practical rule the benchmark enforces: never rank generative reconstruction methods on mean-field error alone, and attach the sampler operating point to every probabilistic number.

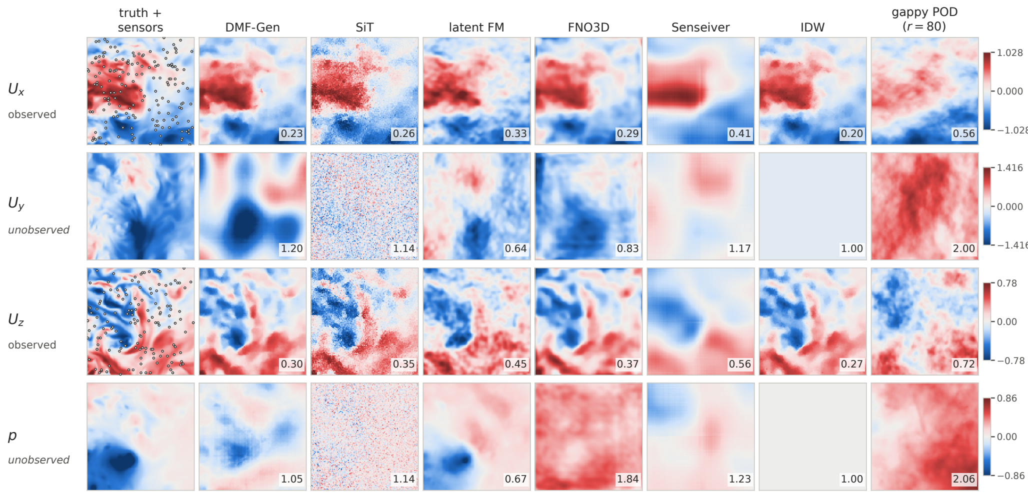

FIG. 6. 3D isotropic turbulence, held-out cube (frame 153), z-midplane. Rows are all four channels: U_x and U_z observed at 19,531 sensors each (1%; in-plane sensors shown as circles), U_y and p never observed; the sensor set is identical in every column. Per-panel numbers are single-sample relative L_2 over the whole cube. The unobserved rows are the identifiability wall of Sec. V B: latent FM, whose decoder imposes global field structure, is the only method below 0.7 on either unobserved channel and places the low-pressure core, while interpolation returns the training mean, gappy POD extrapolates away from the truth, and the point-token sampler emits incoherent noise. CoNFiLD, S3GM (which diverges at this scale), DeepONet++ and Gen4Turb are omitted from the gallery.

TABLE IX. Cost accounting for the 3D-turbulence fleet (H100-SXM 80 GB). Steps are as-run, not equalized: every external baseline received at least DMF-Gen's 48k optimizer steps except SiT-point (wall-clock-matched, 38.5k); CoNFiLD stages train on a wall-clock budget. "(d)" = derived from evaluation-job wall time rather than a dedicated benchmark, and therefore an *upper bound*: it includes data loading, host–device transfer and metric computation, which the dedicated numbers exclude, and per-call setup that a dedicated benchmark amortizes. The two kinds are not interchangeable and we mark rather than mix them; the marked rows are the slow samplers, where the sampling term dominates the overhead and the comparison it supports (orders of magnitude between families) is unaffected. Unifying them would mean re-benchmarking every row on one device, and the H100s these numbers were taken on are no longer the project's hardware, so the honest options are to keep the marked mixture or to re-measure the whole table on the current GPUs; we keep the mixture rather than silently restate the hardware.

Method	Params	Opt. steps	Train s/step	Train GPU-h	Train peak GB	Infer s/field
DMF-Gen (NFE 4)	6.51M	48k	0.574	7.7	53.6	∼17 (d)
Latent flow matching	6.68M	209k+95k	0.238	13.9	3.6	0.019
SiT-point (N=32)	6.57M	38.5k	1.81	19.4	26.8	∼150 (d)
FNO3D (NFE 4)	6.73M	79.9k	0.766	17.0	48.6	0.097
CoNFiLD (pub. prior)	118.9M^{w}	wall	35.5/ep	∼19	4.7	∼130 (d)
Gen4Turb (anneal)	6.16M	105k	0.598	17.4	47.4	1.65
S3GM (JHU-tuned)	6.35M	105k	0.635	18.5	26.3	∼46
Senseiver	6.42M	79.2k	0.776	17.0	22.4	<1
DeepONet	6.51M	82.0k	0.29	6.7	15.7	0.23
IDW k=8 (CPU)	—	—	—	—	10 (RSS)	0.68
Gappy POD r=80 (CPU)	—	45 s fit	—	—	10 (RSS)	0.14

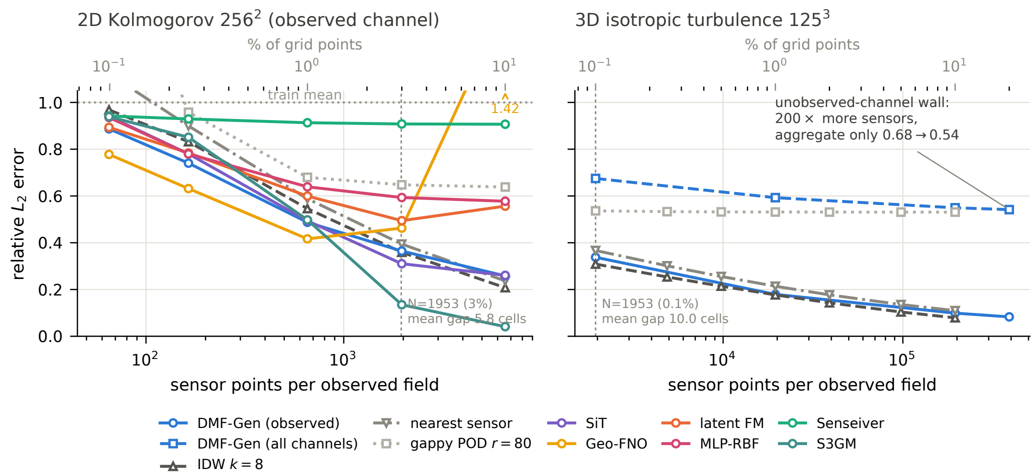

FIG. 7. Reconstruction error versus the *number* of sensor points, in 2D (left, Kolmogorov 256^2, the full learned fleet) and 3D (right, JHU 125^3 cross-cube; solid = observed-channel rel-L_2, dashed = aggregate over all four channels). Colored = learned; gray = classical floors. The two panels sit on opposite sides of the density story: the same sensor count that leaves a 5.8-cell mean gap in 2D leaves a 10-cell gap in 3D, and while every 3D observed-channel error falls steadily with sensor count — DMF-Gen tracking the interpolation slope from 0.33 to 0.09 over 200× — the 3D *aggregate* saturates near 0.54 because the unobserved channels do not identify (Sec. V B). Gappy POD is flat at every 3D density: isotropic turbulence has no low-rank structure for added sensors to project onto. In 2D the fleet fans out rather than sharing a slope: five rows fall monotonically — S3GM steeply enough to pass every other method, interpolation included, from 1965 sensors on — Geo-FNO *degrades* past the canonical density (to 1.42 at 10%, marked off-scale) and latent FM past ∼2000 sensors, and Senseiver is nearly flat throughout. Learned rows: canonical protocol, 50 held-out frames, K as in Table VII, sensor draws identical across methods at every density; classical curves: same seeded sensor draws.

B. Unobserved channels: a fleet-wide identifiability limit

Before the unobserved-channel story, the 2D panel settles a question the single-density tables cannot: *over what range of sensor budgets does learning pay for itself?* Sweeping the Kolmogorov fleet over $\{65, 164, 655, 1965, 6554\}$ sensors — 0.1% to 10% of the grid, canonical draws at every density — bounds that range on both sides for most of the fleet, and shows that the bound is itself method-dependent. At the sparse end learning wins clearly: at 65 sensors Geo-FNO reaches 0.778 and DMF-Gen 0.887 while IDW manages only 0.968 and gappy POD (1.82) and nearest-sensor (1.12) are both worse than predicting the training mean. At the dense end the fleet splits. Plain inverse-distance interpolation reaches 0.208 at 6554 sensors and beats six of the seven learned rows, the best of which are DMF-Gen at 0.258 and SiT at 0.261; for those six the crossover falls between 1965 and 6554 sensors, so their case for learning lives below a few percent sensing — the regime this benchmark targets, and the one uniform-downsampling protocols leave (Sec. V F).

The seventh row inverts that picture. S3GM, the guided video-diffusion prior, is near the bottom of the fleet at 65 sensors (0.939, fifth of seven) and at the top from 1965 sensors on: 0.135 there against SiT's 0.310, and 0.041 at 6554 — five times better than interpolation and six times better than the next learned row. Its curve is the steepest in the benchmark, 23× over the sweep against 3.4× for DMF-Gen, which is consistent with posterior-sampling guidance whose likelihood term strengthens with the measurement count until it overrides the prior; we report that mechanism as consistent with the data, not established by it. The gain has two prices the headline row does not show. It costs 22 s per field against 0.19 for DMF-Gen (Table VII). And S3GM's stated spread does not shrink with its error: spread–error rises from 1.07 at 1% sensing to 2.56 at 10%, and 90% coverage from 0.80 to 0.98, so the most accurate reconstruction in the sweep is also its most over-dispersed. The density conclusion is therefore not that learning stops paying at a few percent sensing; it is that *which* method pays is itself density-dependent. The leader of the single-density ranking in Table VII, Geo-FNO, is last of seven at 10% sensing — worse than predicting the training mean — and the 10% leader is a row the 1% table ranks fourth.

The fleet also does not share a slope, and the exceptions are diagnostic rather than noisy. Five rows fall monotonically over the full 100× range (S3GM 0.939 → 0.041, DMF-Gen 0.887 → 0.258, SiT 0.940 → 0.261, MLP-RBF 0.936 → 0.578, Senseiver 0.942 → 0.907). Two do not: Geo-FNO is best at the canonical density and degrades from there — 0.417 at 655 sensors, 0.463 at 1965 and *1.42* at 6554, worse than predicting the training mean — and latent FM turns the same corner later and more gently (0.495 → 0.557 past 1965 sensors). Both are the rows whose conditioning path resamples the sensors onto a fixed representation — a grid for Geo-FNO, a latent code for latent FM — so adding sensors past the point where that representation saturates adds conditioning mismatch rather than information. This is a failure mode a single-density leaderboard cannot see at all: at the canonical 1% row Geo-FNO is the best learned method on this regime (Table VII), and nothing in that number warns a practitioner that ten times the instrumentation would make it more than three times as bad. Senseiver's near-flatness is the same lesson from the other direction — it extracts essentially all it will ever extract from 65 sensors — and is why we report the density sweep as a first-class result rather than an appendix curve.

On 3D turbulence, no method in the fleet reconstructs the unobserved U_y below 0.60 relative L_2, and most sit near or above 1.0 — worse than predicting the training mean (Table VIII). The two partial exceptions (latent FM at 0.727, CoNFiLD-384d at 0.604) are the two methods whose decoders impose the strongest global field structure, and both pay for it elsewhere (dissipation-range truncation; observed-channel accuracy). This is not a per-method defect: at 1% point sensing on two velocity components, the instantaneous realization of a third component of 3D turbulence is close to unidentifiable, and only its statistics are recoverable. The observation disciplines how such results should be reported — per channel, with unobserved channels flagged — and motivates the cylinder's observe-u-only variant, where the shedding phase *does* determine the unobserved channels and the same test separates methods rather than hitting a physics ceiling. The cylinder's own never-observed pressure channel already makes the contrast quantitative: where 3D turbulence compresses every method into the 0.90–1.00 band, the same measurement on the cylinder spreads them over more than an order of magnitude — gappy POD 0.043, Geo-FNO 0.088, Senseiver 0.095, DMF-Gen 0.128, SiT 0.274, and interpolation at 1.000 by construction (Table V). Identifiability is therefore a property of the flow and the sensing geometry, not a fixed limitation of a method class, and a benchmark that reports only 3D would license the wrong conclusion.

The observe-u-only arm makes the point sharper still. Retraining on u_x sensors alone — so that u_y joins p as an unobserved channel, and the total sensor budget halves from 476 to 238 — leaves Senseiver's u_y error unchanged at 0.231, against 0.240 when u_y was measured directly at 1%. Removing an entire observed channel costs it nothing, because at this Reynolds number the shedding phase inferred from u_x determines the rest of the field; its aggregate moves only from 0.145 to 0.148. DMF-Gen is not indifferent in the same way: its u_y error rises from 0.353 to 0.438 and its aggregate from 0.221 to 0.250. The contrast with Table VIII, where no method recovers an unobserved channel of 3D turbulence at any budget, is the cleanest statement the benchmark makes about identifiability: whether a channel is recoverable is set by the flow, and *how efficiently a method exploits that recoverability* is what separates architectures. Reporting only the aggregate would hide both halves.

C. Calibration: characterized, then repaired

Almost every scored ensemble in the fleet is under-dispersed at the headline operating point (spread–error 0.37–0.94 against the ideal 1.0; coverage 0.33–0.63 against 0.90), consistent with every scored precedent in adjacent literatures.[15,16] The two exceptions are instructive rather than contradictory: cylinder latent FM is over-dispersed (spread–error 1.26, coverage 0.89) and Kolmogorov S3GM sits just past unity (1.07, coverage 0.80). Both are latent- or guidance-driven samplers whose spread is set by a mechanism other than the observation density, and neither is more accurate for it — latent FM's relative L_2 on the cylinder (0.192) is twice the Geo-FNO row's. Calibration and accuracy are separately earned, which is the case for reporting both. For the best-characterized model the miscalibration has a low-dimensional mechanism: predictive spread follows a distance-to-nearest-sensor profile with a hard floor ($\approx$0.09 in standardized units), the profile is density-invariant, and the sampler step count rescales it wholesale ($\approx$1.5$\times$ from NFE 4 to 16). That structure makes post-hoc repair tractable, and the benchmark ships two estimators fitted strictly on the tuning split and frozen on test: a per-density scalar spread multiplier, and conformalized quantile scaling binned by distance-to-nearest-sensor, which carries a finite-sample marginal coverage guarantee on exchangeable points. The identical wrapper is offered to the generative baselines, so the repaired comparison is not one-sided. Three honesty requirements govern the reporting: the repair is post hoc and stated as such; it rescales stated uncertainty and cannot sharpen the spread floor, which remains the residual limitation; and all repaired numbers carry their operating point.

The scalar estimator, fitted on the tuning half and evaluated frozen on the disjoint test half, moves the test spread–error ratio to within a few percent of unity wherever the miscalibration is stationary. On 3D turbulence the DMF-Gen row is repaired at every density — $0.436 \to 1.001$ at 0.1% sensing (α=2.30), $0.550 \to 1.017$ at 1% (α=1.85), and $0.949 \to 1.016$ at 10% (α=1.07) — and the multiplier falls toward one as sensing densifies, exactly as the distance-to-sensor mechanism predicts. The same wrapper applied to latent FM on the identical protocol gives $0.706 \to 0.993$, and on Kolmogorov it repairs all four generative rows ($0.674 \to 0.975$ for DMF-Gen, $0.691 \to 0.985$ SiT, $0.729 \to 0.981$ latent FM). It also corrects in the other direction: the over-dispersed Kolmogorov S3GM row is shrunk rather than inflated (α=0.95, $1.046 \to 0.996$), which is the behaviour a calibration repair should have and which an inflation-only heuristic would miss.

The cylinder is the honest counterexample and we report it as one. There the repair does not transfer: DMF-Gen overshoots ($0.582 \to 1.112$) and latent FM is essentially untouched (α=1.01, $1.189 \to 1.200$), because the held-out set spans two Reynolds numbers whose error and spread scales differ, so a single scalar fitted on one half is the wrong functional form for the other. A scalar repair assumes the miscalibration is stationary across the reporting set; when the held-out set is heterogeneous by construction, that assumption fails and the fix must be conditioned on the regime variable. This is a limitation of the estimator, not of the diagnostic, and it is why we report the fit rule, the split, and the before/after pair for every row rather than a single headline ratio. (The cylinder's canonical frames are all even-indexed, so its tune/test halves are taken by position in the frame list rather than by absolute index parity; both halves still draw from both held-out Reynolds numbers, and the rule used is recorded in each output JSON.) The conformalized estimator repairs the quantity the scalar one cannot. Fitting a per-distance-bin conformal quantile on the tuning snapshots and freezing it on test, 90% coverage on U_x moves from 0.403 to 0.895 at 0.1% sensing, 0.560 to 0.900 at 1%, and 0.755 to 0.904 at 10%, with the unobserved channels behaving the same way (0.409 to 0.898 on p); the same holds at every nominal level we report. That is the finite-sample split-conformal guarantee doing exactly what it promises, on five million held-out points. The two estimators also agree about how much repair the model needs: the raw coverage deficit shrinks monotonically with sensing density (0.40, 0.56, 0.76) exactly as the scalar multiplier falls toward one (2.30, 1.85, 1.07), which is the signature of both instruments measuring a single underlying miscalibration rather than two unrelated defects.

One estimator caveat is worth recording, since the benchmark's stance is that estimators must be reported with their failure modes. The conformal fit bins by distance to the nearest sensor, and at 10% sensing that distribution concentrates enough that equal-count bin edges coincide and a bin can come up empty on the tuning half. An empty bin has no quantile; emitting zero there would produce zero-width intervals and destroy coverage for exactly the points that land in it. We collapse duplicate edges and fall back to the pooled quantile for any empty bin, which is conservative rather than optimistic, and the reported bin count is the post-collapse one.

Reporting both estimators together is more informative than either alone, because they disagree in a diagnostic way. After the conformal repair the *coverage* is exact but the spread–error ratio overshoots to 1.2–1.6, while the scalar multiplier sets spread–error to 1.0 and leaves coverage short. Neither is wrong: a scalar matches the second moment, a conformal quantile matches the tail probability, and the two coincide only for Gaussian residuals. The gap between them is therefore a measurement of how non-Gaussian the reconstruction error is — heavy-tailed enough that the interval achieving 90% coverage is half again wider than the standard deviation would suggest. A benchmark that reported a single calibration number would hide that, and a practitioner who needs honest intervals should take the conformal one while a practitioner who needs an error bar on the mean should take the scalar. [TODO: same treatment for the latent-FM dumps, so the calibration comparison is two-sided as the scalar one already is.]

D. Spectral fidelity splits by band

Windowed shell spectra of single samples (Fig. 8) separate the generative families where pointwise error cannot. On 3D turbulence the latent route holds more inertial-range energy than the ambient point-cloud route (0.62 vs. 0.41 of DNS energy on the observed U_x, $k \in [8, 31]$) but collapses in the dissipation band, retaining 0.07–0.09 of DNS energy over $k \in [32, 45]$ against the ambient model's 0.40–0.46 — the autoencoder spectral floor[25,26] appearing where it was predicted, a factor-of-five separation invisible to relative L_2. Neither family reproduces the DNS spectrum from 1% sensors: the ambient model's high-k content flattens into a broadband-noise plateau on the unobserved pressure channel (3.5–4.8× DNS energy), and on channels no sensor constrains its samples stay near its spectrally empty source prior (inertial ratio 0.03–0.06 on U_y, versus 0.67 for the latent decoder that paints texture everywhere). Member-level spectra, windowed estimation, and per-channel reporting are all load-bearing; ensemble-mean spectra and unwindowed estimators each silently manufacture different conclusions.

E. Computational cost and scaling

Cost separates the fleet along two axes. At the benchmark resolution (Table IX), total training cost to a finished model spans 7–19 GPU-hours across learned methods — comparable — but inference spans five orders of magnitude, from 19 ms (latent decode) to minutes (chunked token generation, DPS optimization), and peak training memory spans 3.6–54 GB. Against resolution (Fig. 9), the fleet splits into two families: grid computation (voxel diffusion, convolutional autoencoders) scales cubically with measured out-of-memory walls at 320^3 (training, batch 1) and 640^3, while point computation (attention over sensors with chunked queries) is flat to 1024^3 in both training and inference memory. The honest converse is that grid computation is cheaper below the crossover ($\sim 240^3$ for inference: a convolutional decode of a 125^3 field costs 6 ms where chunked point evaluation costs seconds). No family dominates; the benchmark's contribution is that the walls and crossovers are measured, so a practitioner can place their discretization on the axis before choosing a method family.

Figure 10 makes the same point from the accuracy side: on the cost–accuracy plane

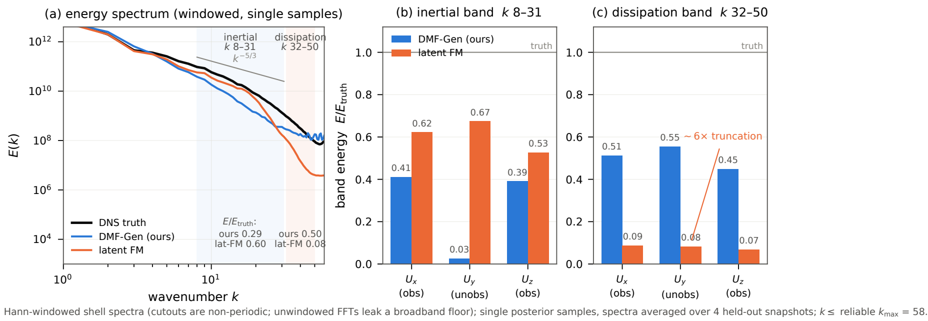

FIG. 8. Turbulence statistics of single posterior samples on the held-out 3D region (4 snapshots, NFE 4, Hann-windowed; ensemble means are deliberately not shown — averaging destroys small scales by construction). Left: shell-averaged energy spectra — the latent baseline tracks the DNS through the inertial range then falls away sharply in the dissipation band (autoencoder floor); the ambient model sits lower in the inertial range and flattens into a noise plateau at high k. Middle: longitudinal structure functions. Right: velocity-gradient PDFs.

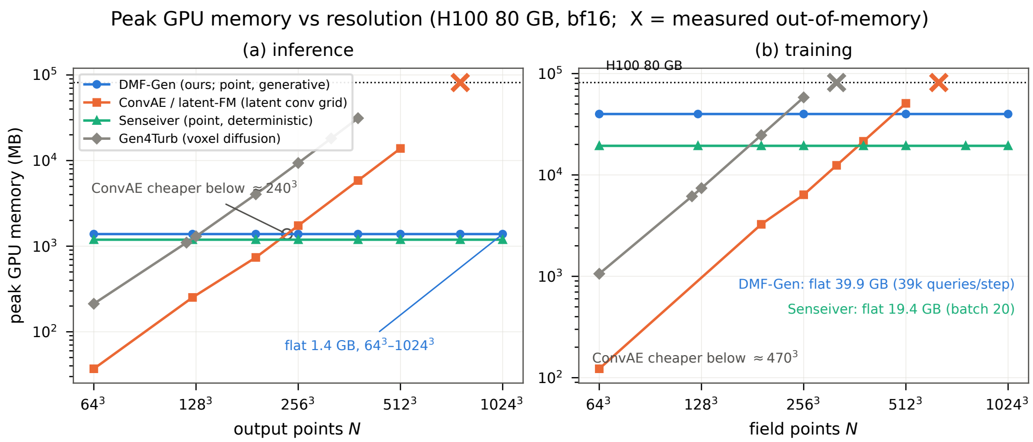

FIG. 9. Measured peak memory versus output resolution (H100, bf16, log–log; OOM markers are measured failures, not extrapolations) for full-field sampling (left) and training steps (right). Grid methods (voxel diffusion, convolutional AE) scale cubically and fail at 320^3 and 640^3; point methods (DMF-Gen, Senseiver) are flat to 1024^3 (10^9 points). Annotation: fraction of the field supervised per training step — 100% forced for grid methods, a fixed 39k-point budget for the subsampled point method.

alone the latent baseline dominates every method including DMF-Gen, and the case for ambient point-cloud generation rests entirely on the axes the plane omits — resolution scaling, discretization freedom and dissipation-range fidelity, measured here, and operator robustness, which the deferred multiphysics regime is designed to measure (Sec. VI).

F. Why benchmark numbers look worse than the literature's — protocol, not architecture

Readers who compare our tables and galleries against published reconstruction papers — including papers evaluated on the very same Kolmogorov data[8] — will notice that every method here, including the strong ones, scores and *looks* worse. We consider this observation a central result of the benchmark rather than an embarrassment, because each contribution to the gap is a protocol choice, and we measure them separately.

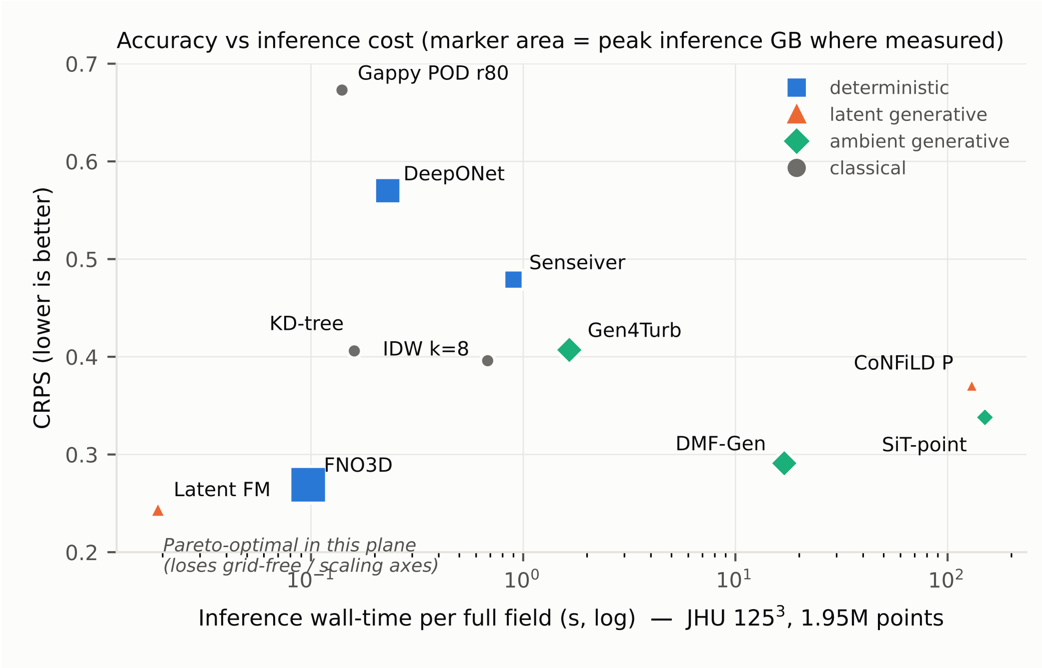

FIG. 10. Probabilistic accuracy (CRPS, JHU cross-cube, n=50) against inference wall-time per full 125^3 field, marker area proportional to peak inference memory where measured (Table IX). At the benchmark resolution the latent flow-matching baseline Pareto-dominates this plane outright — fastest and lowest CRPS — which is precisely why a single cost–accuracy plot is an insufficient basis for method choice: the axes it cannot show (Fig. 1; the scaling walls of Fig. 9; the dissipation-band ratios of Sec. V D) are where the grid-locked methods lose. S3GM is absent from this plane: its upstream sampler diverges at 3D scale, and its two labelled retuned arms are reported in Table VIII rather than plotted as if they were the published method (Sec. IV D).

(i) Splits — severe in 3D, and, we were surprised to find, negligible in 2D. On the 617 consecutive DNS frames of the 3D cutout, frame-to-frame correlation is 1.00 at unit separation and 0.67 at separation 100, and a shuffled random-in-time split places every evaluation frame within one frame of a training frame. Under that split the latent flow-matching baseline scores 0.14 relative L_2; under the spatially disjoint protocol the same architecture, data, and budget scores 0.56, and the fleet shifts by factors of 2–4 *unevenly*, so leakage there does not merely inflate scores, it reorders methods.

We expected the 2D benchmark to reproduce this and it does not. We re-evaluated frozen checkpoints on frames drawn from *training* trajectories, positioned two raw frames from a training frame, where the measured field correlation to the nearest training frame is $r = 0.97$ (raw-frame correlation on this dataset: 0.992 / 0.972 / 0.916 at lags 1 / 2 / 4). Despite near-duplicate evaluation targets, no method gained materially: DMF-Gen 0.487 → 0.483, SiT 0.494 → 0.491, Senseiver 0.913 → 0.873, against a training-free inverse-distance control that moved 0.546 → 0.537 and therefore sets a ≈1.6% frame-difficulty floor. Net of that control, the largest memorization benefit in the fleet is under three percent.

The distinguishing variable is plausibly data scale rather than temporal proximity: the 2D training set holds 2560 snapshots against the 3D protocol's 150, and at that ratio these ∼6.5M-parameter models apparently learn the sensing operator rather than a lookup table, so a near-duplicate target buys them almost nothing. We report this as a measured contrast with a hypothesis attached, not a demonstrated mechanism. Its practical consequence is the useful part, and it is a caution against the easy version of our own argument: *leaky splits are not automatically fatal — they are fatal in the small-data regime where the 3D sparse-reconstruction literature actually operates*, which is precisely where random-in-time splits remain standard. The pre-fix 3D evaluations we archived were run on a single snapshot at a different solver setting, so we quote the latent-flow-matching figure that is directly comparable and do not tabulate the remainder.

That result relocates the blame. Of the total gap to literature-style reporting, the sensor *layout* carries almost all of it: moving from scattered to uniform-lattice sensing takes DMF-Gen from 0.487 to 0.367, a 25% reduction, and combining uniform sensing with the leaky split adds essentially nothing beyond it (0.364). The interpolation floor moves the same way (IDW 0.546 scattered, 0.452 uniform), which is the tell that this is a property of the measurement geometry rather than of any model: uniformly spaced sensors bound the worst-case distance to data, and reconstruction error at 1% sparsity is dominated by that distance. A benchmark that adopts uniform sensing because it is convenient therefore reports a systematically easier problem, and no amount of split hygiene compensates for it. Our headline claim is consequently narrower than we first drew it: *protocol, not architecture* remains right, but within protocol it is the sensing layout that dominates, and the split matters mainly as a correctness requirement rather than as a large numerical effect. [TODO: PROVENANCE: the 3D pair above (0.14 shuffled / 0.56 honest) comes from archived pre-fix evaluations that are not on the current host; per the text they were single-snapshot at a different solver setting. The honest 0.56 must therefore not be read against Table VIII, whose latent-FM row is 0.469 over 50 snapshots (0.56 is that row's *coverage*); state the archived setting beside the pair. The "factors of 2–4" range has no tabulated source in the artifact set.]

(ii) What "1% sparsity" means. Much of the visually striking literature conditions on uniformly downsampled coarse grids — every cell of a coarse lattice, i.e. band-limited full-field information with bounded gaps — or adds physics guidance (PDE-residual or forcing terms at sampling time[8,10]). Our protocol uses randomly scattered points and purely data-driven conditioning: the same nominal percentage carries far less information and admits large sensor-free regions. Measured on Kolmogorov with the same checkpoint and frames, rearranging the 655 scattered sensors onto a uniform 25×26 lattice improves DMF-Gen from 0.487 to 0.367 relative L_2 (CRPS 0.235→0.171) and IDW from 0.546 to 0.452 — a 25% accuracy difference from sensor *layout* alone, before any change of model or split. Layout dominates: combining the uniform lattice *and* the seen-trajectory split gives 0.364, statistically indistinguishable from the lattice alone, so at 2D data scales essentially the whole protocol gap that we can reproduce is sensor geometry rather than leakage.

(iii) What is shown. Publication figures typically display ensemble means or favorable frames; we display single posterior samples on fixed, pre-registered frames (on the Kolmogorov gallery frame the four leading rows score 0.57–0.82 as single samples, against 0.42–0.50 for the same rows' fifty-frame ensemble-mean averages), because single-sample behavior is what the physics metrics score. Ensemble size interacts with the same point: K=8 coverage understates calibration (Sec. II E), and ensemble means visually erase exactly the small scales the spectra measure.

(iv) Matched budgets. Every model trains at a matched ∼50k-step budget with disclosed tuning asymmetry, not to per-method convergence. Fairness costs polish; the cost table prices it.

The synthesis we intend a reader to take away: *apparent reconstruction quality in this literature is substantially a property of evaluation protocol, and only secondarily of architecture.* A benchmark that fixes the protocol first makes the remaining — genuine — architectural differences measurable.

VI. DISCUSSION

a. Which method should a practitioner use? The benchmark's answer is conditional, and that is the point. If sensing is dense relative to the flow's intrinsic dimension (laminar, periodic regimes) use gappy POD: on the cylinder it reaches 0.056 aggregate rel-L_2 against 0.093 for the best learned method, recovers the unobserved pressure channel, and costs a least-squares solve against a rank-20 basis (Sec. IV A). On observed channels of turbulence at ∼1% density, training-free interpolation remains competitive with the best learned methods — the case for learning begins where the problem is ill-posed: unobserved channels and sparser sensing in this suite, and, beyond it, corrupted or surface-confined operators. There, deterministic learners are structurally limited to conditional means; among generative routes, latent compression buys inference speed and global field structure at a measured dissipation-range floor in 3D, while ambient point-cloud generation buys 3D small-scale fidelity at higher inference cost — an ordering that inverts on Kolmogorov (Sec. IV C) — with geometry-freedom and operator amortization as the structural advantages the deferred regimes are designed to test. If the discretization exceeds $\sim 300^3$, grid families exit on measured walls; if it lives on a body-fitted or unstructured mesh, they exit on interface grounds, as the cylinder's grid export already illustrates. Uncertainty from any current method must be recalibrated before operational use, and the benchmark ships the wrapper.

b. Limitations. All three regimes are simulation data on canonical flows, and all three are Cartesian or near-Cartesian — two periodic Cartesian grids and one body-fitted O-grid around a circular cylinder — so the present suite measures neither measurement realism nor geometric generality: the multiphysics regime with realistic corrupted measurement operators (wildfire LES) and the unstructured, case-varying geometry regime with surface-only sensing (transonic wing family) are deferred to a follow-up paper, and the corresponding claims about operator robustness and geometry-freedom are stated here only as structural properties of the interfaces, not as measured results. All learned rows train per-regime (no pretraining across flows); the suite contains no experimental data — particle-image-velocimetry or pressure-sensitive-paint regimes with real sensor noise are the natural next addition; single-snapshot reconstruction ignores temporal information that assimilation exploits; the cost accounting is single-GPU (H100) and does not probe multi-GPU scaling; the tuning-budget asymmetry of Sec. II F item (v) is disclosed but not eliminated; and training-seed variance is measured for the 2D fleets (Sec. IV C) but not for the 3D one, where a single replicate costs more than the rest of the suite combined.

VII. CONCLUSION

FlowRecon-Bench turns sparse-sensor flow reconstruction from a collection of incompatible per-paper evaluations into a measured landscape: three regimes spanning 2D laminar to 3D chaotic flow, twelve-plus audited methods, proper probabilistic scoring, physics metrics with their estimator pitfalls documented, matched cost accounting, and leakage-controlled splits with the distortion of the dominant protocol quantified. The findings that outlive any single table: method ranking is regime-dependent and no method wins twice; the classical floor is sometimes unbeaten, and which classical method constitutes it changes with the regime; unobserved channels are near-unidentifiable in 3D turbulence yet recoverable on the cylinder by a linear basis, so identifiability is a property of the flow rather than of the method; mean-field error alone misranks generative methods, and on Kolmogorov it reverses the ranking between accuracy and CRPS across whole method families; nearly every scored ensemble is under-dispersed and post-hoc recalibration repairs stated coverage; and the computational axis separates grid from point families at measured walls. The datasets, splits, sensor manifests, configurations, audit notes, and evaluation driver are released for extension — the leaderboard is meant to be beaten.

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DATA AVAILABILITY

The benchmark artifact — datasets or regeneration recipes (JHTDB cutout coordinates and download scripts, OpenFOAM case definitions, converter scripts), split manifests, seeded sensor draws, per-method configurations, audit notes, and the evaluation driver — will be released on acceptance. [TODO: Zenodo/HF DOI; confirm JHTDB redistribution terms (fallback: recipe + coordinates); needs sign-off before anything goes public.]

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